Class-5 BA MODEL Date..................

* The Random network model differs from Real networks in two ways:-

1) Growth

→ Random model assumes NO' of Nodes {N} is fixed & network doesnt evolve over time

→ In Real Networks, Nodes are continuously Added & the Network size keeps growing.

2) Preferential Attachment

→ Random Network - connections are formed Randomly.

→ Real Network - New Nodes prefer connecting to already well- connected nodes.

$$\underbrace{\text{Growth} + \text{Preferential Attachment}}_{\text{BA Model}} \Rightarrow \text{Scale Free Networks} \left\{ \begin{array}{l} \text{Power law} \\ \text{with Hubs} \end{array} \right\}$$

Date..................

* Degree Distribution of BA Model

$$\boxed{P(K) \sim K^{-\gamma}, \quad \gamma = 3} \rightarrow \text{Scale free Property}$$

Degree growth of a Node in BA

$$k_i(t) = m\left(\frac{t}{t_i}\right)^{\beta}$$

$k_i(t)$:- degree of node i at time t

t_i :- time when node i joined

m :- number of links added per Node.

Beta = 1/2

→ The degree exponent ($\gamma = 3$) is independent of m.

→ The degree dist. of BA model is independent of time & of the system size N.

∴ Network reaches Statitionary Scale free state

→ The coefficient of powerlaw dist. is proportional to m^2

∴ m affects scale, not shape of the Network.

i.e More links → Denser Network → but still scale free.

Date..................

* Absence of Growth & Preferential Attachment

1) **Model A** :- Growth ✓ , Preferential Attachment ✗

Growth ✓ ↓ new Nodes Added

Preferential Attachment ✗ ↓ links are uniform | Random

Attachment probability ($\pi(k_i)$) → constant

We get a degree distribution :- $P(K) \sim e^{-K}$ which is Exponential

So Exponential Distribution

2) **Model B** :- Growth ✗ , Preferential Attachment ✓

Growth ✗ ↓ fixed no. of Nodes

Degree Distribution

- Initially looks like Power law
- Then becomes Gaussian
- Eventually - Fully connected Network.

★ We Need both Growth + Preferential Attachment to get a true scale free Network.

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Date..................

* BA Model Summary

1) Number of Nodes = time (growth steps)

$$\boxed{N = t}$$

2) **Number of links**

$$\boxed{L = mt}$$ {each new Node adds m links}

3) **Average Degree**

$$\boxed{\langle K \rangle = 2m}$$

3) **Degree Dynamics**

$$\boxed{k_i(t) = m\left(\frac{t}{t_i}\right)^{\frac{1}{2}}}$$ ∴ older Nodes → Higher degree

4) **Degree Distribution**

Power law :- $P(K) \sim K^{-3}$, Exponent $\gamma = 3$

This is the scale free property.

5) Diameter

$$\boxed{\langle d \rangle \sim \frac{\log N}{\log \log N}}$$

→ ultra small world due to Hubs

6) clustering

$$\boxed{\langle c \rangle \sim \frac{(\ln N)^2}{N}}$$

→ Goes to 0 as N ↑↑ {low clustering as Network grows}.

* Limitations of the BA Model

1) Fixed Exponent

BA Model predicts $\gamma = 3$ while Real networks have $2 \le \gamma \le 3$

2) undirected only

BA → undirected, Real Networks (www) → directed.

3) No Real world processes

Node deletion, link deletion, rewiring.

4) Does not allow to distinguish b/w nodes

∴ BA Model → Minimal, Proof of Principle model
only Explains why Scale free emerges
& not real networks.